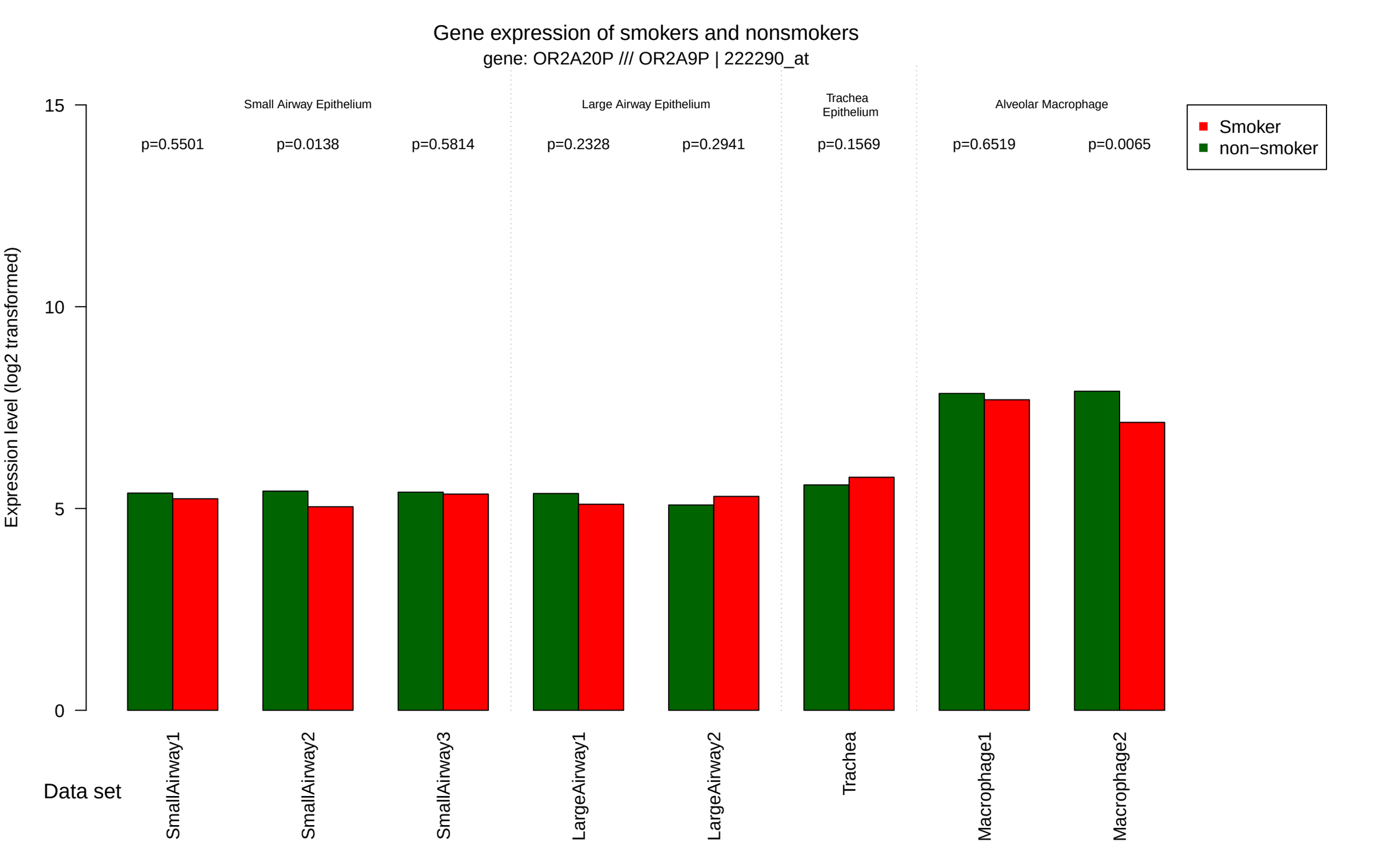
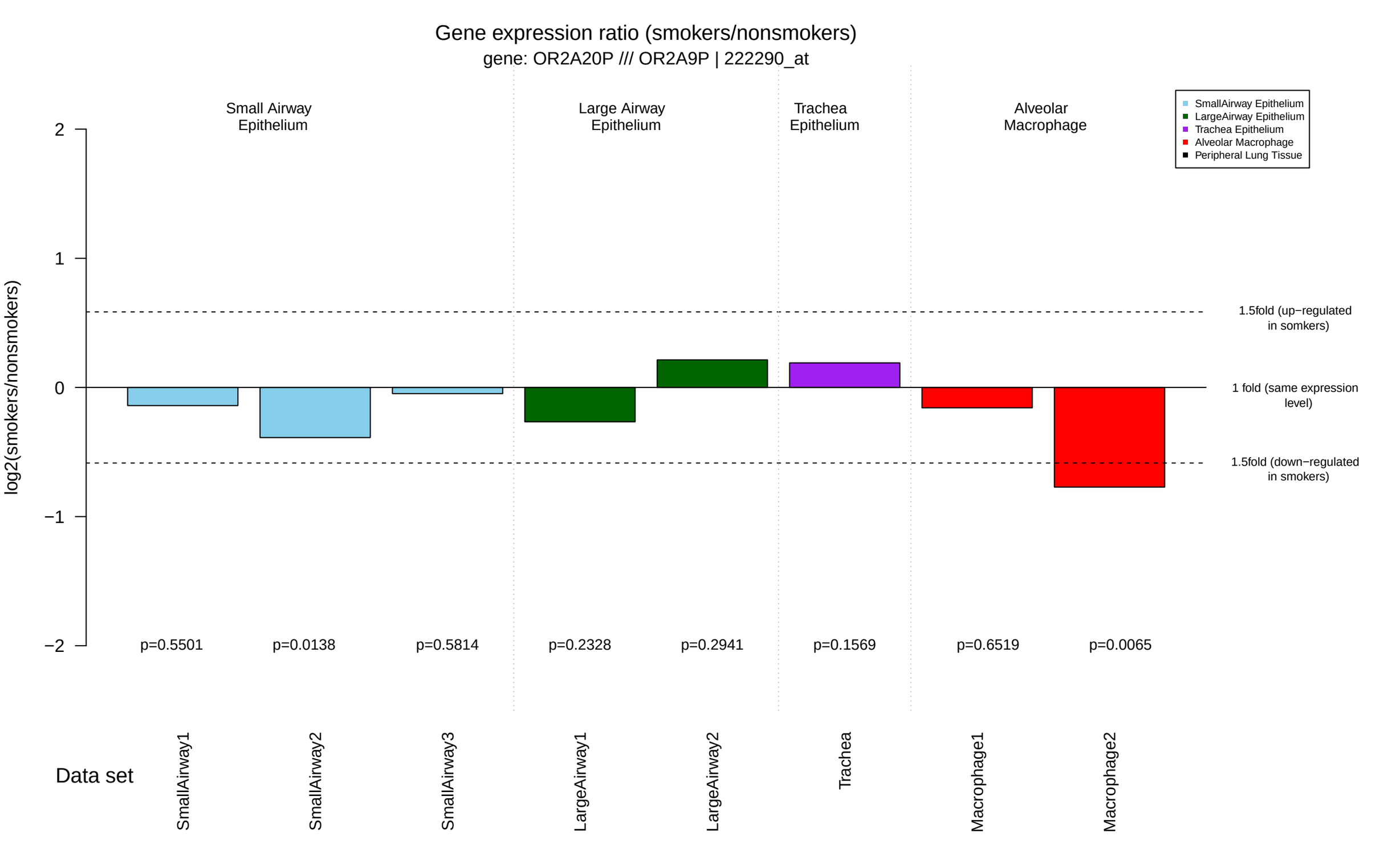
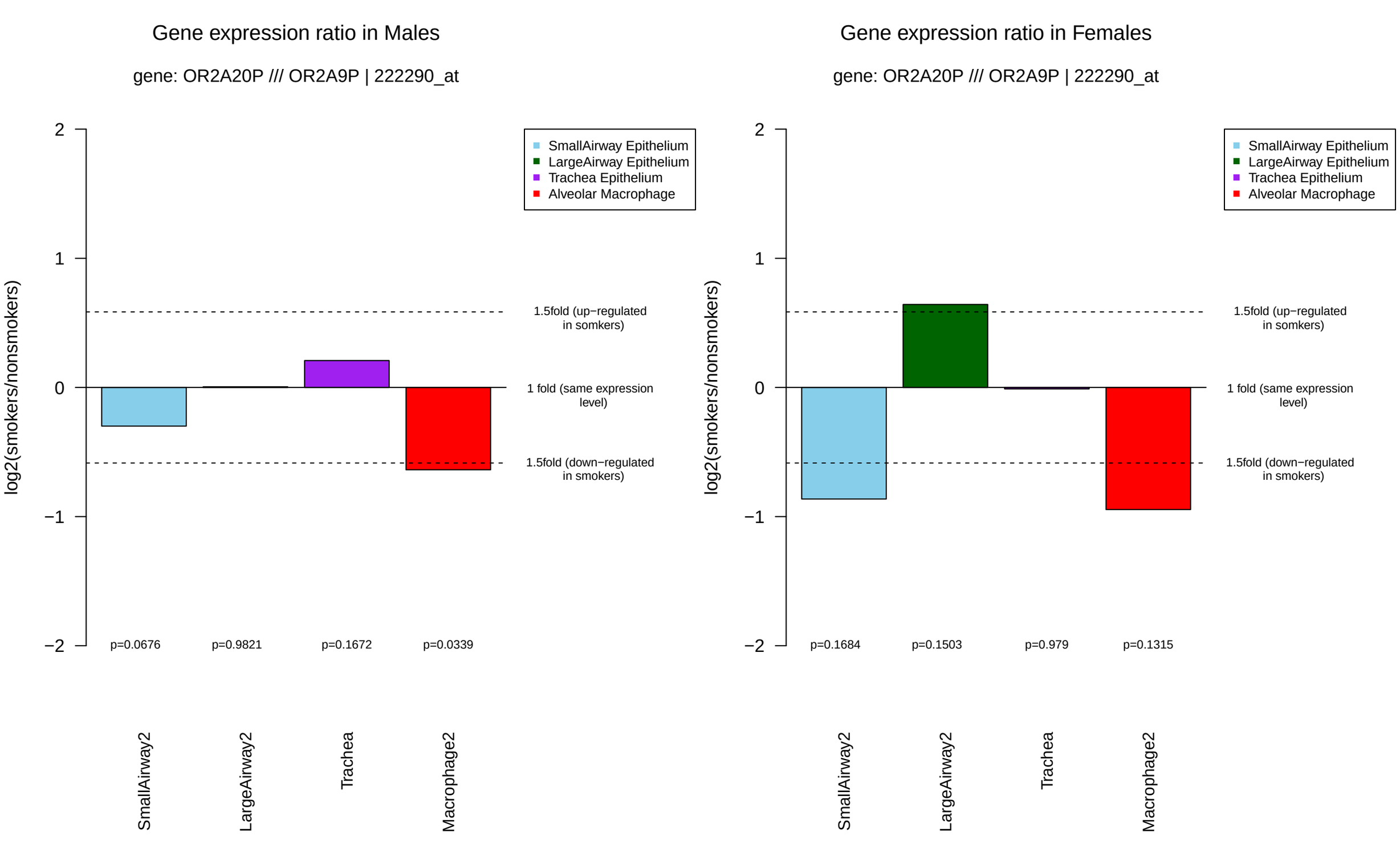
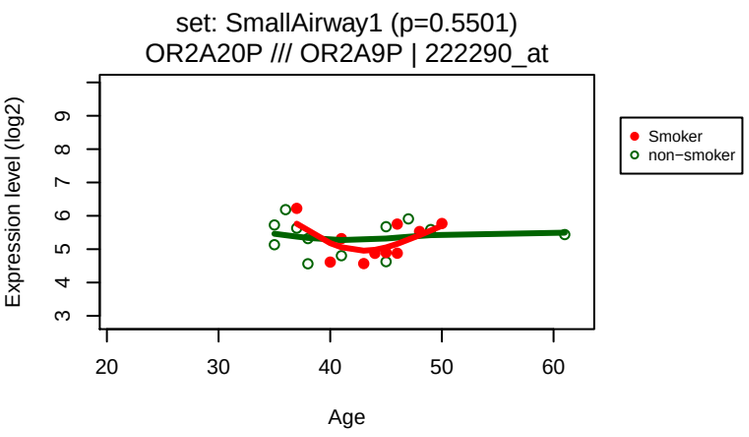
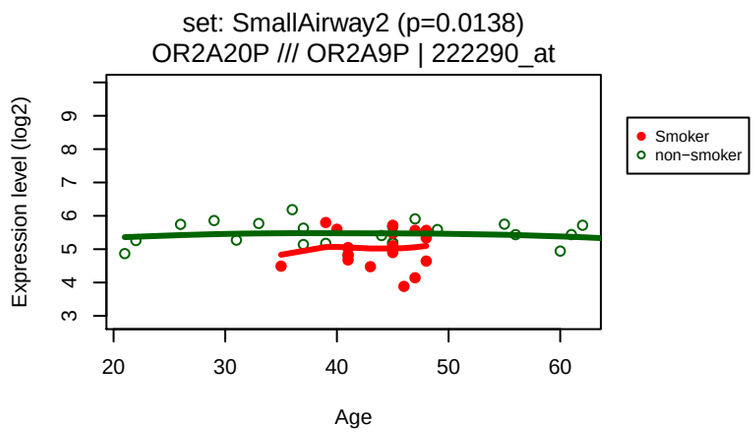
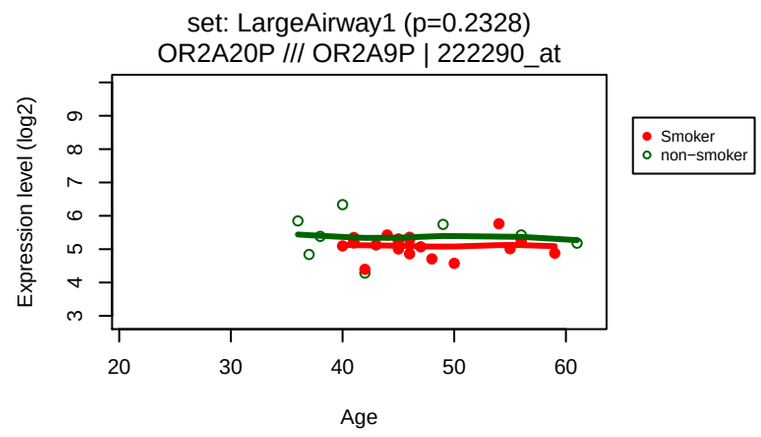

Gene expression of smokers and nonsmokers
gene: OR2A20P /// OR2A9P | 222290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5501
p=0.0138
p=0.5814
p=0.2328
p=0.2941
p=0.1569
p=0.6519
p=0.0065
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

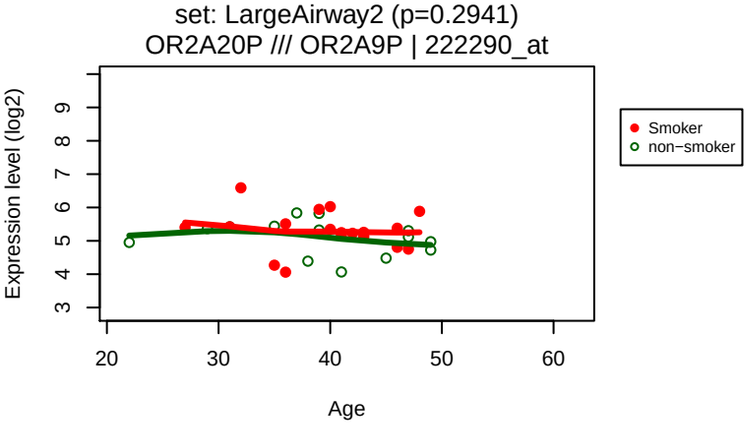

Gene expression ratio (smokers/nonsmokers)
gene: OR2A20P /// OR2A9P | 222290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5501
p=0.0138
p=0.5814
p=0.2328
p=0.2941
p=0.1569
p=0.6519
p=0.0065
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

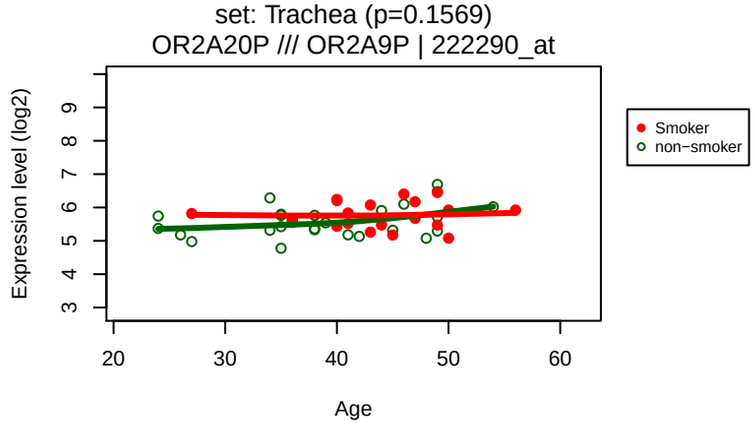

Gene expression ratio in Males
gene: OR2A20P /// OR2A9P | 222290_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0676
p=0.9821
p=0.1672
p=0.0339
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2A20P /// OR2A9P | 222290_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1684
p=0.1503
p=0.979
p=0.1315
SmallAirway2
LargeAirway2
Trachea
Macrophage2

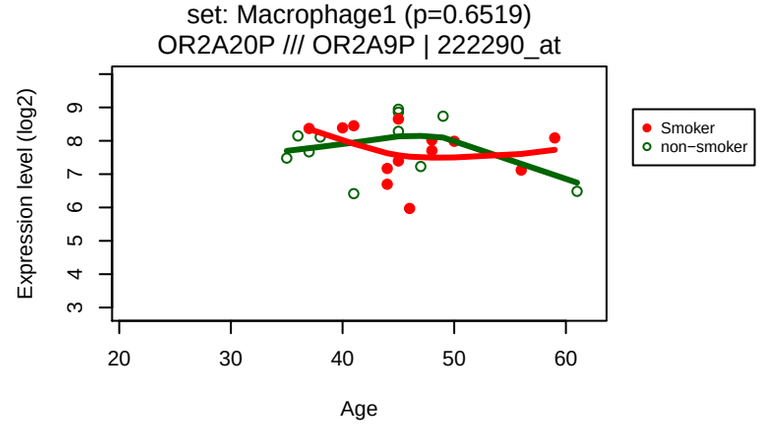

set: SmallAirway1 (p=0.5501)
OR2A20P /// OR2A9P | 222290_at
Expression level (log2)
Age
Smoker
non−smoker

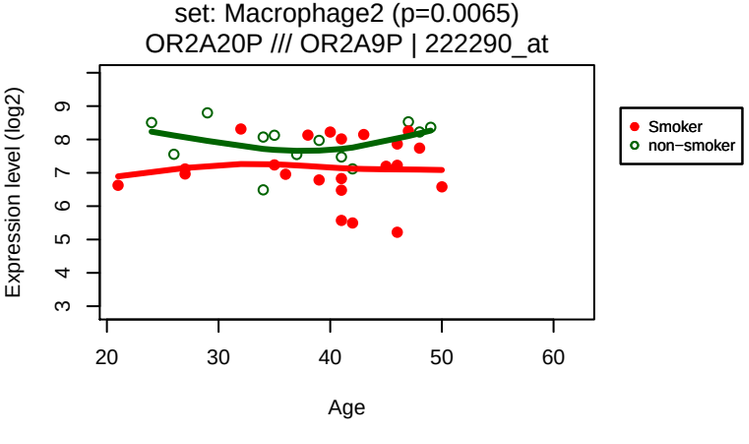

set: SmallAirway2 (p=0.0138)
OR2A20P /// OR2A9P | 222290_at
Expression level (log2)
Age
Smoker
non−smoker

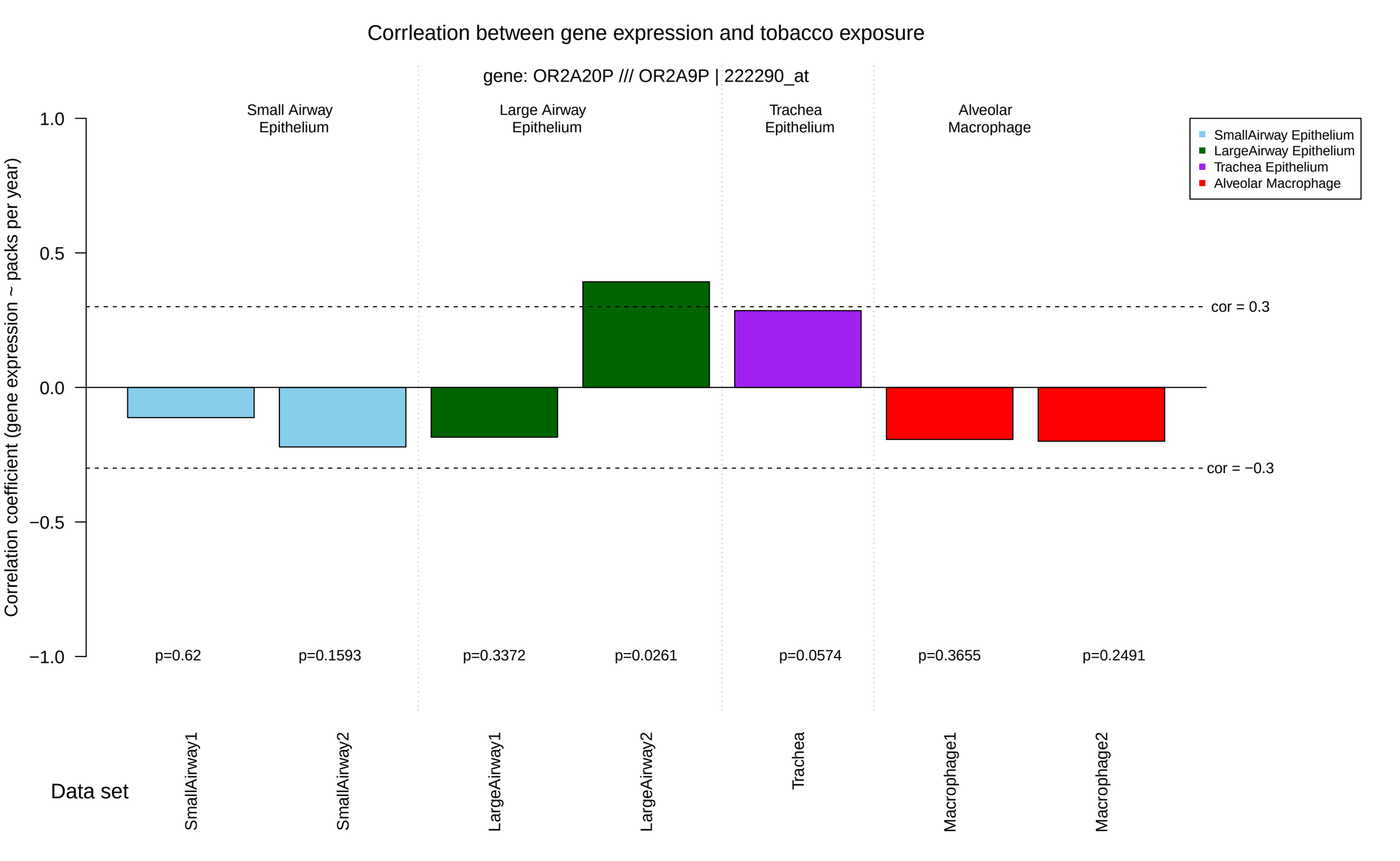

set: LargeAirway1 (p=0.2328)
OR2A20P /// OR2A9P | 222290_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2941)
OR2A20P /// OR2A9P | 222290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1569)
OR2A20P /// OR2A9P | 222290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6519)
OR2A20P /// OR2A9P | 222290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0065)
OR2A20P /// OR2A9P | 222290_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2A20P /// OR2A9P | 222290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.62
p=0.1593
p=0.3372
p=0.0261
p=0.0574
p=0.3655
p=0.2491
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2